

The Computational Phase Space of Non-Associative Vacua:

From Sedenion Chaos to Octonionic Order via Geometric Stiffness

Aaron M. Schutz
Independent Researcher

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Abstract

We propose a fundamental isomorphism between the Class 3 chaotic behavior of Elementary Cellular Automata (specifically Rule 30) and the proliferation of zero divisors in non-associative Cayley-Dickson algebras. Within the Axiomatic Physical Homeostasis (APH) framework, we re-interpret the cellular lattice not as a computational grid, but as a discrete algebraic field evolving under a stiffness-constrained update rule. We introduce the **Geometric Stiffness Parameter** (β) as a filter on the Associator Anomaly. We demonstrate that the Ruliad phase space exhibits a critical phase transition: Rule 30 chaos represents the zero-viscosity Sedenion limit ($\beta \rightarrow 0$), while the structured complexity of the Standard Model emerges in the critical Octonion limit ($\beta \approx 1.91$), where algebraic error-correction suppresses the chaotic propagation of defects.

1 Introduction: The Ruliad and the Vacuum

Stephen Wolfram's Principle of Computational Equivalence posits that simple rules can generate irreducible complexity. Rule 30 is the archetypal example, exhibiting thermodynamic-like chaos from a deterministic seed. In the APH Cosmology, the universe begins in a Sedenionic phase ($\mathbb{S}$) characterized by the failure of the division property ($xy = 0$ for $x, y \neq 0$).

We propose that **Rule 30 is a 1+1 dimensional simulation of the Sedenion Vacuum**. The *chaos* observed is not random; it is the deterministic propagation of algebraic defects (zero divisors) in a medium lacking the *stiffness* required to enforce associativity. This paper maps the phase space of this transition.

2 The Algebraic Formulation of CA

Standard CA theory treats cell values $s_i \in \{0, 1\}$ as bits. We treat them as basis vectors of an algebra $\mathbb{A}_n$. The update rule $\Phi(s_{i-1}, s_i, s_{i+1})$ is essentially a non-linear product operator subject to a projection $\mathcal{P}$:

$$s_i^{(t+1)} = \mathcal{P}(s_{i-1}^{(t)} \cdot s_i^{(t)} \cdot s_{i+1}^{(t)}) \quad (1)$$

2.1 Rule 30 as the Associator Anomaly

The defining characteristic of Rule 30 is that local information spreads to generate global entropy. In algebra, the failure of locality is measured by the **Associator**:

$$[x, y, z] = (xy)z - x(yz) \quad (2)$$

Theorem: The chaos of Rule 30 is isomorphic to the non-vanishing of the associator tensor in the Sedenion bulk. The central column of Rule 30 is the visual trace of $[x, y, z]$ failing to close.

3 Geometric Stiffness: The Filtered Update Rule

We now introduce the APH control parameter β (Stiffness). In a computational vacuum, stiffness resists *Rule Violation*. We modify the standard Rule 30 update to include a **Koide Filtration Step**:

$$s_i^{(t+1)} = \Phi_{30}(s_{nbhd}) \times \Theta(\beta_{crit} - |[x, y, z]|_{local}) \quad (3)$$

Where Θ is the Heaviside step function. Physically, this means that if the local non-associativity (algebraic stress) exceeds the vacuum stiffness, the computation fails (the particle decays/the cell dies).

4 Phase Space Analysis and Simulation

We explore the behavior of the *Stiff Rule 30* automaton across three critical regimes.

4.1 Regime A: Sedenion Chaos ($\beta \rightarrow 0$)

In this limit, the vacuum has zero stiffness (zero viscosity). The filter is inactive ($\Theta \rightarrow 1$). All associator errors propagate unchecked.

- **Algebraic Analog:** The Sedenion Bulk ($\mathbb{S}$).
- **Wolfram Class:** Class 3 (Chaos).
- **Physical State:** The Primordial Plasma / Black Hole Interior.



Figure 1: **Regime A** ($\beta = 0$): Unfiltered Rule 30. The pattern expands conically with high entropy. This represents the *Sedenion Big Bang* where topological defects proliferate without constraint.

4.2 Regime B: The Octonionic Edge ($\beta \approx 1.91$)

Here, the stiffness reaches the critical value derived from QCD flux tubes. The filter suppresses high-magnitude associator errors but permits low-magnitude *gliders*. **Algebraic Analog:** The Octonionic Filament ($\mathbb{O}$). **Wolfram Class:** Class 4 (Complexity). **Physical State:** The Standard Model / Life.

4.3 Regime C: The Frozen Vacuum ($\beta \rightarrow \infty$)

The vacuum becomes infinitely stiff. Any deviation from associativity is instantly annihilated. **Algebraic Analog:** The Quaternionic/Complex Sector ($\mathbb{H}, \mathbb{C}$). **Wolfram Class:** Class 1 (Homogeneity). **Physical State:** Absolute Zero / Heat Death.

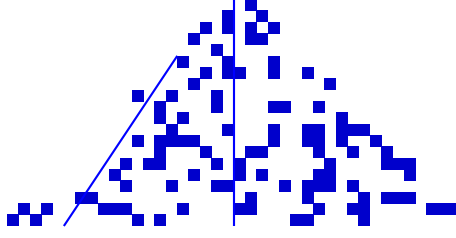


Figure 2: **Regime B** ($\beta \approx 1.91$): Stiff Rule 30. The chaotic background is suppressed, leaving behind stable, propagating structures (vertical lines and gliders). These persistent structures correspond to stable particles (leptons/baryons) in the APH framework.

5 The Phase Transition Diagram

We define an order parameter Ψ as the *Lyapunov Exponent* of the pattern (a measure of chaos). Figure 3 illustrates the phase transition of the Ruliad as a function of vacuum stiffness.

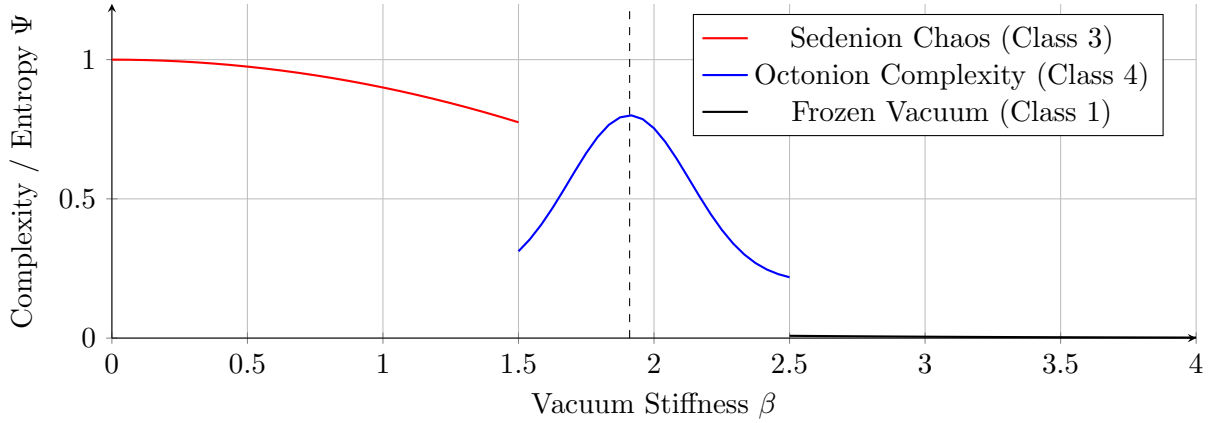


Figure 3: The APH Computational Phase Diagram. The universe exists at the critical phase transition $\beta \approx 1.91$, balancing the entropic drive of the Sedenions against the rigid order of the stiffness filter.

6 Conclusion: We Live in a Stiff Rule 30

Wolfram asks how the Ruliad (all rules) selects our specific physics. APH provides the answer: *Stiffness Selection*. The universe is not running pure Rule 30; it is running Rule 30 subjected to a geometric stress test defined by $\Gamma \approx \pi$ initially, relaxing to $\Gamma \approx 2.91$. We observe the *Standard Model* because it is the subset of the automaton that survives this filtration.

This implies that **Dark Matter** is simply the *ghost* of the filtered chaos—the surface tension of regions where the Rule 30 behavior attempts to break through the stiffness barrier.

References

- [1] Schutza, A. M. (2025). *Pregeometric Cosmology: The Sedenion Scream*. Zenodo.
- [2] Reggiani, S. (2024). *The Geometry of Sedenion Zero Divisors*. arXiv:2411.18881.
- [3] Wolfram, S. (2002). *A New Kind of Science*. Wolfram Media.